Ivan **Cucchi**

PhD student

About me

I'm driven to advance AI techniques to foster innovation and deeper insights in medical and pharmaceutical research.

Areas of specialization

Machine Learning • Protein Design • Computational Mathematics • Computational Chemistry • Tutoring

Personal

Nationality: Italian
Birthday: 15/02/1992
Residence: Stockholm (Sweden)

Languages

Italian C2 mother tongue
English C1

Social

Contact

ivanqkk92@gmail.com
ivanqkk92@naiss.se
ivan.cucchi@dbb.su.se
ivan.cucchi@scilifelab.se
+39 3341686128
sites.google.com/ivan-cucchi

Curriculum

2026–present **Application specialist in AI for life science**
Stockholm University · Stockholm
As an Application Specialist in AI for Life Science, I work at the interface of SciLifeLab, NAISS, and the MIMER AI competence center, supporting life science researchers in the development and deployment of advanced AI and machine-learning solutions on national high-performance computing infrastructures. My responsibilities include curating and managing large, heterogeneous biological datasets; assisting with applications for computational resources; and designing end-to-end machine learning pipelines for applications such as microscopy image analysis, biomolecular structure prediction, genomics, and precision medicine. I also contribute to user support, training activities, and workshops, and collaborate with national and international research infrastructures to promote the effective adoption of AI methodologies in life science research.

2023–2026 **PhD student**
University of Pavia · Pavia
PhD student in Computational Mathematics, Learning, and Data Science, enrolled in the thematic program titled *High Performance Computing and Deep Learning Methods for Protein Evolution and Design*. Research involves the analysis and prediction of protein stability, folding, and function, using data from molecular dynamics simulations and structural bioinformatics. Work includes designing hybrid AI models that combine sequence-based and structural features, with the goal of supporting drug discovery and therapeutic protein engineering. The project is carried out in collaboration with the University of Pavia and Università della Svizzera italiana (USI) in Lugano, with results published in peer-reviewed scientific journals.

2025–2026 **PhD visiting student**
SciLifeLab · Stockholm
3-month visiting period at SciLifeLab as part of my PhD program. I have regularly attended meeting and conferences, actively cooperating with Prof. Elofsson and his research team.

2025 **Mathematics Instructor**
Istituto d'Istruzione Superiore Simone Weil · Treviglio
10-hour targeted mathematics course focused on student skill potentiation and reinforcement of core concepts.

2024 **PhD visiting student**
Università della Svizzera Italiana · Lugano
6-month visiting period at Euler Institute of Università della Svizzera Italiana as part of my PhD program. I have regularly attended meeting, conferences and courses, actively cooperating with Prof. Krause and his research team.

2022 **Intern**
Sync Lab S.r.l. · Milan
Development of integration flows using TIBCO BusinessWorks, through lessons, practical exercises, and real enterprise use cases, covering software engineering, distributed systems, relational databases with SQL, and implementation of projects with TIBCO BW and BPM as part of a postgraduate internship in the engineering and design area.

2021–2022 **Trainer**
Ver Italia — Turoting & Education · Monza
Provided educational support through private tutoring for students, focusing on study methods and school activities.

Education

2019–2022 **Master in Mathematics**
University of Pavia · Pavia

2011–2019 **Bachelor in Mathematics**
University of Pavia · Pavia

2007–2011 **Graduation in Scientific High School**
Niccolò Copernico · Pavia

Thesis

2022 **Machine Learning Methods for Identifying Atrial Fibrillation in 12-lead ECGs**
Supervisor: Luca F. Pavarino · University of Pavia
The work focused on identifying atrial fibrillation through the analysis of 12-lead ECG signals using supervised Machine Learning techniques. Explored and compared the performance of k-Nearest Neighbors, Random Forest, and Multilayer Perceptron classifiers. The study involved preprocessing clinical data, implementing classification models, and evaluating their accuracy in detecting cardiac arrhythmias. Integrated concepts from physiology, signal processing, and computational modeling to enhance understanding and support preventive diagnostics in cardiology.

2019 **Numerical Models and Cardiac Electrophysiology Simulations for Long QT Syndrome**
Supervisor: Luca F. Pavarino · University of Pavia
The work focused on modeling and simulating Long QT Syndrome (LQTS) through numerical representations of cardiac electrophysiology. Explored and implemented biophysical models such as Hodgkin-Huxley, Luo-Rudy, and O'Hara-Rudy to reproduce action potentials and ion current dynamics. Simulations were conducted to study the electrical behavior of cardiac cells under normal and pathological conditions. The study integrated concepts from cardiac physiology, mathematical modeling, and computational simulation to deepen the understanding of arrhythmogenic mechanisms and support research in cardiology.

Publications

2025 **Cucchi, I.**, Frasnetti, E., Frigerio, F., Cinquini, F., Pavoni, S., Pavarino, L. F., & Colombo, G. (2025). *MOLECULE: Molecular-dynamics and Optimized deep Learning for Entropy-regularized Classification and Uncertainty-aware Ligand Evaluation*. Journal of Chemical Theory and Computation, 21(18), 9186–9199.

2025 Mansoor, S., Frasnetti, E., **Cucchi, I.**, Magni, A., Bonollo, G., Serapian, S. A., Pavarino, L. F., & Colombo, G. (2025). *LARGE SCALE ENERGY DECOMPOSITION FOR THE ANALYSIS OF PROTEIN STABILITY*. Cell Stress and Chaperones, 30(1), 57–68.

2024 Frasnetti, E., **Cucchi, I.**, Pavoni, S., Frigerio, F., Cinquini, F., Serapian, S. A., Pavarino, L. F., & Colombo, G. (2024). *Integrating Molecular Dynamics and Machine Learning Algorithms to Predict the Functional Profile of Kinase Ligands*. Journal of Chemical Theory and Computation, 20(20), 9209–9229.

Academic Supervision

2025 **Integrating Molecular Dynamics Simulations with Machine Learning and Deep Learning to Predict Nanobody Binding Modes**
Candidate: Mattia Lai · Master Thesis · University of Pavia
Supervisor: Luca F. Pavarino and Giorgio Colombo
Co-Supervisor: Gauthier Trèves and Ivan Cucchi
The work focused on predicting nanobody–antigen binding mechanisms by integrating molecular dynamics simulations with supervised machine learning approaches. Physicochemical descriptors were extracted from high-quality crystallographic structures and used to train and compare machine learning and deep learning classifiers, optimizing the trade-off between predictive performance and computational efficiency. The models were developed in a target-free setting, without prior assumptions on binding sites or antigen type, and addressed both orthosteric and allosteric recognition mechanisms. Feature importance analysis was then performed to uncover the structural determinants distinguishing the two binding classes, transforming machine learning from a purely predictive tool into a physics-informed framework for molecular insight in drug discovery.

Talks

Sep 2025	[*Talk*] **5th Edition of Young Applied Mathematicians Conference**. 26 September 2025, University of Padua.
May 2025	[*Seminar*] **Caffè Beltrami**. 26 May 2025, University of Pavia.
May 2025	[*Poster*] **Spring Workshop COMPMAT**. 23 May 2025, University of Pavia.
May 2025	[*Poster*] **ICAM Workshop**. 22 May 2025, University of Pavia.
Apr 2025	[*Poster*] **1st Workshop on Data Science for Health and Biology**. 7 April 2025, Health Data Science Centre at Human Technopole & Politecnico di Milano.
Jan 2025	[*Talk*] **3rd Workshop of UMI Group — Mathematics for Artificial Intelligence and Machine Learning**. 27–31 January 2025, University of Bari.
Jun 2024	[*Poster*] **Scientific Machine Learning, emerging topics**. 18–21 June 2024, SISSA — International School for Advanced Studies.
May 2024	[*Talk + Poster*] **Spring Workshop COMPMAT**. 8 May 2024, University of Pavia.

Academic Activities — Assistant Teaching

2024–2025	**507290 — Programming 2** BSc in Mathematics · University of Pavia
2024–2025	**507289 — Programming 1** BSc in Mathematics · University of Pavia
2024–2025	**500142 — Mathematics** BSc in Economics · University of Pavia
2023–2025	**502886 — Dynamical systems: theory and numerical methods** MSc in Mathematics and Bioengineering · University of Pavia

Programming

- Python
- MATLAB
- Julia
- LaTeX

Modules

- Numpy
- pandas
- PyTorch
- RDKit
- scikit-learn
- SciPy

Volunteering

2016–2023 **Rescuer**
Croce Bianca Milano · Binasco
Volunteered as a rescuer with Croce Bianca Milano, Binasco section, providing emergency medical assistance, patient transport, and support during urgent interventions, demonstrating teamwork, stress management, and a strong commitment to public service.

Certificates & Grants

2024	B2 English
2023	Fundamentals of Accelerated Computing with CUDA Python
2023	NVIDIA DLI Certificate – Accelerating End-to-End Data Science Workflows
2023	NVIDIA DLI Certificate – Getting Started with Deep Learning
2022	PNRR-MUR Research Programme CN00000013 *National Centre for HPC, Big Data and Quantum Computing*
2009	Shorei-Kan Karate Gojuryu Sho Dan